

More Gauss's Law — the Three Symmetries

PHYS 235 · Wed Sep 23 · Knight 24.4–24.6

Name: _____

Goal. State Gauss's law and understand what it says, then use it to *compute* $\vec{E}$ — not just check it — for the distributions where symmetry makes that possible: a point charge and a shell, an infinite line, an infinite sheet, and a uniformly charged sphere.

1. What Gauss's law says

a) Flux. The number of electric field lines that exit or enter a _____ is the same, no matter the _____ of the surface. By “number of field lines” we mean *flux*:

$$\Phi = \underline{\hspace{2cm}}$$

Notes: the surface's area vector is _____ to the surface and points _____, by convention. If lines are exiting, Φ is _____; if entering, Φ is _____. A line that just passes through counts +1 for leaving and −1 for entering, so it contributes _____ net.

b) Relation to charge. The number of field lines — also called _____ — through a closed surface is equal to _____. Does it matter how far from the charge the surface is? _____ The closed surface we integrate over is called a _____.

Check. A point charge sits at the center of a Gaussian sphere. If you double the sphere's radius, does Φ change? Does E at the surface change? (Different questions!)

2. Using it: match the surface to the symmetry

a) Use Gauss's law when the charge distribution has one of the three classical symmetries: _____, _____, _____.

Definition of symmetry:

b) Choose a Gaussian surface that respects the _____ of the charge distribution. If the distribution is spherically symmetric, make your surface spherically symmetric too, _____ (similarly for cylindrical and planar symmetry).

If you've done this, $\vec{E}$ will be everywhere either _____ or _____ to your Gaussian surface, and $\Phi = E \oint dA = EA$ — Gauss's law becomes plain algebra: $EA = q_{\text{enc}}/\epsilon_0$.

Check. Why does a cube fail as a Gaussian surface for a single point charge?

3. Warm-up: a point charge, then a shell

Sketch a point charge $+q$ with a Gaussian sphere of radius r around it.

By symmetry $\vec{E}$ is radial and constant in magnitude on the sphere, so $EA = q_{\text{enc}}/\varepsilon_0$ with $A = 4\pi r^2$ and $q_{\text{enc}} = q$ gives $E = \underline{\hspace{2cm}}$ — Coulomb's law, recovered from Gauss's law.

Now make it a thin **shell** of charge Q , radius R , instead of a point. **Outside** ($r > R$): the Gaussian sphere encloses $\underline{\hspace{2cm}}$, so $E = \underline{\hspace{2cm}}$ (identical to a point charge — the shell might as well not have a size). **Inside** ($r < R$): the Gaussian sphere encloses $\underline{\hspace{2cm}}$, so $E = \underline{\hspace{2cm}}$.

Check. The sphere you'll work below is *solid*, not a shell — charge fills the whole volume. Will its field inside behave the same way as the shell's?

4. Infinite line of charge

Linear charge density λ . $\vec{E}$ points radially out from the wire and cannot depend on position along it, so the Gaussian surface is a *coaxial cylinder* of radius r and length L .

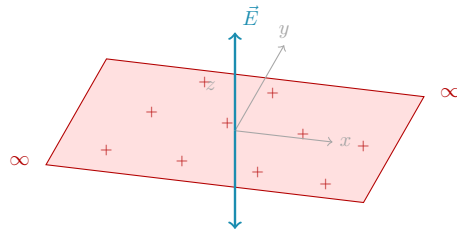
Sketch the wire and a coaxial Gaussian cylinder of radius r and length L . Label both, and the field arrows.

Which part of the cylinder carries flux (and which carries none)? Area A , then q_{enc} , then solve for E . Show L cancelling.

E for an infinite line

5. Infinite sheet of charge

Surface charge density η . This is a *plane*, not a line: it extends forever in *two* directions, x and y , and $\vec{E}$ points straight away from it along $\pm z$ on both sides.



An infinite sheet, seen at an angle: it extends forever in x and y , and $\vec{E}$ points straight out along z on either side.

Now look at it *edge-on* (sighting along y , so the sheet appears as a line and the pillbox is easy to draw), with end caps of area A_{cap} :

Sketch the sheet edge-on, with a pillbox straddling it. Label the half-height, the cap area A_{cap} , and the field arrows.

Total area carrying flux, q_{enc} in terms of η and A_{cap} , then E .

E for an infinite sheet

Check. Two surprises. A_{cap} cancelled — why did it *have* to? And the answer contains no r at all. What does that say about walking away from the sheet?

Then: two parallel sheets, one $+\eta$ and one $-\eta$. Superpose the two fields you just found — nothing new to derive.

Sketch two parallel sheets, $+\eta$ and $-\eta$. Write $\vec{E}$ in each of the three regions: above, between, below.

6. Uniformly charged sphere

Total charge Q spread uniformly through a solid sphere of radius R . There are two regions, and the only difference between them is q_{enc} .

Sketch the solid sphere of radius R , and a Gaussian sphere of radius r — draw it once outside ($r > R$) and once inside ($r < R$).

Outside ($r > R$): the Gaussian sphere encloses _____ of the charge.

q_{enc} , then E . What familiar result does this match?

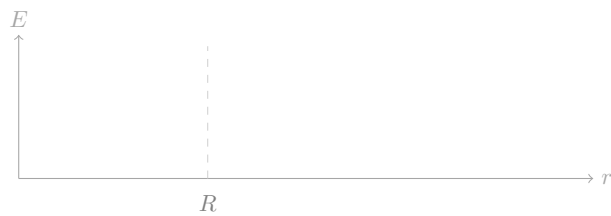
E outside the sphere

Inside ($r < R$): only the charge within radius r counts. The charge fills the _____, so the fraction enclosed is the ratio of _____.

$q_{\text{enc}} = ?$ then $E = ?$

E inside the sphere

Check. Sketch E against r across both regions (use the axes below). At $r = R$ — jump, kink, or neither?



All three results are in the *Applying Gauss's Law* sim — open its **Info** panel to check yours.